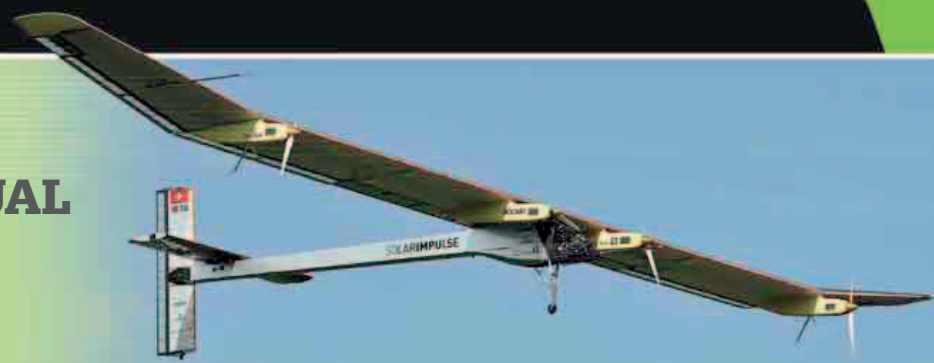


“WE ARE ON THE VERGE OF PERPETUAL FLIGHT.”

Bertrand Piccard, Swiss co-founder of the Solar Impulse project, 2010



The Solar Impulse used 2,200 sq ft (200 sq m) of solar panels to generate electricity to power the electric motors that drove its propellers.

originally found in 1983, in a disused quarry in Germany; Hurum came across it in 2006 and was intrigued by features that made it stand apart from the lower primates—including humanlike nails and opposable big toes. The species was named *Darwinius masillae*, in honor of British naturalist and pioneer of the theory of evolution Charles Darwin (see 1859). Later that year, in October, scientists announced that the bones of a 4.4-million-year-old hominid fossil, *Ardipithecus ramidus*, were **the oldest hominid remains ever discovered**.

Meanwhile, two projects began to increase knowledge of our own species. In October 2008, the **Human Microbiome Project** was launched. Spearheaded by the US National Institutes of Health, this was an initiative to investigate the **microbes that colonize different parts of the body** and establish the roles of these organisms in health and disease. A year later, the **Human Epigenome Project**, an international project to map the epigenome (factors that affect which genes are switched on and when) published its **first map of the human epigenome**.

IN 2010, IBM AND INTEL BEGAN MANUFACTURING CHIPS with 32-nanometer transistors. Semiconductor manufacturers had been cramming ever more transistors on to a single chip since the invention of integrated circuits (see 1958). Advances like these helped increase the power and portability and decrease the cost of various electronic devices, and the first commercially successful tablet



Synthetic mycoplasma bacteria
This colored micrograph shows the first organisms with a synthetic genome. Nicknamed “Synthia,” the new species is officially called *Mycoplasma mycoides JCVI-syn1.0*.

computer, **Apple’s iPad**, was released in 2010. It was soon followed by tablet computers running the Android operating system, which had been developed by a consortium headed by Google, and had its first release in 2008. Android was already a popular operating system for smartphones.

There is a limit to how small conventional transistors can be, so researchers were looking for alternatives to silicon as the basis of future electronics. In February 2010, IBM researchers created the **first reliable, fast-switching transistor** from the material graphene (see 2004). In July 2009, scientists at ETH Zurich, Switzerland, had created a transistor made from a single molecule, and in May 2010 a team at the University of New South Wales, Australia, made a transistor with just seven atoms, using phosphorus.

In May, a team led by US biologist **Craig Venter** (see panel, right) created the **first synthetic life form**. Using a technique called oligonucleotide synthesis—in which DNA sequences stored in a computer are pieced together to order—Venter and his co-workers had already created a complete virus genome (2003) and a synthetic chromosome (2007). In 2010, they synthesized a copy of the **genome of a bacterium called *Mycoplasma mycoides***, made a

CRAIG VENTER (b.1946)

US biologist Craig Venter is a pioneer in genome studies and synthetic biology. In 1998, he helped form Celera Genomics, a private company that helped speed up sequencing of the human genome (see 2000). In 2006, he founded the J. Craig Venter Institute in California, a centre of expertise in genomics that produced the first synthetic life form.



few changes (including adding a kind of “watermark”), then inserted it into another bacterial cell that had had its DNA removed. The new genome functioned as required, manufacturing proteins and causing the cell to reproduce. Biologists hope to learn a great deal about living systems by building them themselves, rather than just taking them apart. Synthetic life also holds the promise of designing **new life forms that could be beneficial**

—for example, by helping clean up oil spills or produce biofuels.

In July, a solar-powered aircraft called **Solar Impulse made a continuous 26-hour flight**. Electricity generated by solar panels was stored in batteries during the day, allowing the aircraft to fly through the night. The project was the brainchild of Swiss balloonist Bertrand Piccard (b.1958) and Swiss entrepreneur André Borschberg (b.1958), and it aimed to encourage development of renewable energy. Later in the year, US company SpaceX launched Dragon, the **first commercially owned spacecraft to go into orbit**. In 2012, it completed the first of a number of scheduled missions delivering supplies to the International Space Station.

1 MILLION

THE NUMBER OF **DNA BASE PAIRS** USED TO MAKE THE **FIRST ORGANISM WITH A SYNTHETIC GENOME**

October 1, 2009
Scientists announce that fossil remains of the oldest hominid remains ever discovered

October 2009
Scientists publish the first map of the human epigenome

February 2010
Researchers at IBM create the first reliable fast-switching graphene transistor

March 20, 2010 First full face transplant performed in Spain

May 2010 Australian physicists make a transistor from just seven phosphorus atoms

May 21, 2010 Scientists create the first synthetic life form, a microorganism with an artificially created genome

April 3, 2010 Apple releases the iPad, the first successful tablet computer

May 7, 2010 Scientists announce completion of the sequencing of the Neanderthal genome

July 8, 2010 Solar Impulse becomes the first solar-powered aircraft to complete a non-stop 24-hour flight

September 29, 2010 Astronomers discover Gliese 581g, a possible candidate for a habitable extrasolar planet

December 8, 2010 US company SpaceX launches the first commercially owned spacecraft to reach orbit